

Jimmies and Joes or X's and O's?

More Risk for Less Return in Modern NBA Shot-Selection

An ML and Markowitz Portfolio Analysis of 2nd-Order Shot Statistics in a 1st-Order Analytic League

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1. Introduction

The first-wave of basketball analytics has focused almost exclusively on first-order statistics. But first-order results are often *average*, obscuring deviance. We should *expect* more. By first-order logic, the *St. Petersburg Paradox* presents unboundedly-high ROI for any finite entry-fee (Bernoulli, 1738). A first-order maximizer accepts the bet at any earthly entry-fee. Yet, most won't commit their lunch money for entry (Hayden and Platt, 2009). Risk also matters in NBA basketball despite the League's bounded-ROI. Game-level scoring-risk is substantial, and should be second-order only statistically. NBA teams compete for game-wins, not season-table-wins, given a statistically-moderate number of game-possession; scoring-variance is substantial and directly-linked to shooting-ability and shot-selection, the latter a short-run choice-variable.

2. Methods

Shot-portfolio variance is a point-value-weighted sum of binomial-variances for each FGA and FGA-pursuant FTA, both linked to the original FGA-type/zone-location, and **shot-portfolio return** follows from the same inputs:

$$\sigma^2 = n \sum_{z=1}^Z \text{prop}_z \cdot \left[\text{Pts}_z^2 \cdot \text{FG}\%_z \cdot (1 - \text{FG}\%_z) + \text{FT rate}_z \cdot \text{FT}\% \cdot (1 - \text{FT}\%) \right], \quad (1)$$

$$\text{Pts}/\text{FGA}_z = \text{Pts}_z \cdot \text{FG}\%_z + \text{FT rate}_z \cdot \text{FT}\%, \quad \text{PPG} = n \sum_{z=1}^Z \text{prop}_z \cdot \text{Pts}/\text{FGA}_z. \quad (2)$$

where z is shot-type/zone-location, n is FGAs per team-game, prop_z the type-conditional shot-frequency, $\text{Pts}_z \in \{2, 3\}$ is attempt point-value, $\text{FG}\%_z$ the zone-conditional conversion-rate, FT rate_z the zone-conditional shot-pursuant FT-rate, and $\text{FT}\%$ FT-accuracy, so that $(\text{FT rate}_z \cdot \text{FT}\%)$ is zone-conditional made FT-rate. We treat NBA team-game shot-selection as a portfolio of heterogeneous Bernoulli-trials differing in zone-distribution and thus in points-expectation, scoring-risk/variance, and shot-pursuant FT-rate, where unfouled 3PA (2PA) variance scales by 9 (4). Using modern, **Markowitz portfolio-theory** from finance, **step-linear programming**, and **GAM-ML modeling**, we model first- and second-order shot-selection statistics to determine how *expected* are game-level expectations. **Each shot-portfolio** has a **quantifiable risk-return signature**. The risk-return efficient shot-selection frontier solves a step-linear program. Risk is the “bad” input and scoring the “good” output. Both moments depend on the cross-zone attempt-share vector, $\text{prop} = (\text{prop}_1, \dots, \text{prop}_Z)$. A portfolio's quality is the ordered-pair $\text{RiskReturn}(\text{prop}) = (\sigma(\text{prop}), \text{PPG}(\text{prop})) \in \mathbb{R}_+^2$ in the risk-return Cartesian-plane, and season- t league-frontier is the attainable set's upper-envelope for a league-representative team,

$$F_t(s) = \max_{\text{prop} \in \Delta^{Z-1}} \{ \text{PPG}_t(\text{prop}) : \sigma_t(\text{prop}) \leq s \}, \quad \Delta^{Z-1} = \left\{ \text{prop} \in \mathbb{R}_+^Z : \sum_{z=1}^Z \text{prop}_z = 1 \right\}. \quad (3)$$

The **Markowitz-frontier** allows for no attainable pair to its northwest. Partitioning the offensive-floor into six zones over 390 team-seasons from 2013-14—2025-26, the 30-stadium tracking-era, a GAM-ML model learns the efficiency-frontier for each season by inputting 2,340 zone/team/season trios, fitting **attempt-weighted, ridge-penalized cubic-splines smoothed** in season, team FT-accuracy (exogenous shooting-skill), and pace. A box-score identity yields each shot-type's shot-pursuant FT-scoring value. Accordingly, **Fig.1** visualizes the **~2.77-million FGAs** from 2013-26.

Two counterfactuals, one at a time: the surface is the frontier, the line is the actual path

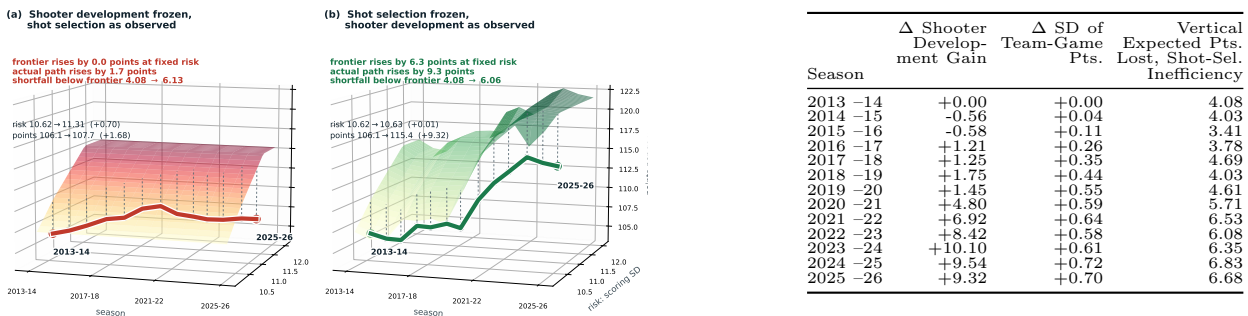
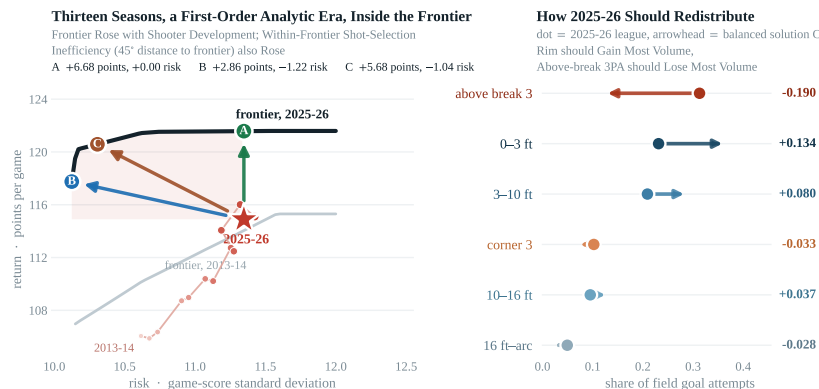


Figure 1: For each plot of Fig.1, the surface is the frontier, and the line is the actual path. Left panel: Shooter development frozen at 2013-14 with shot selection as observed over time. Right panel: Shot selection frozen with shooter development as observed over time. The frontiers show that offensive shooting efficiency gains were driven by shooter-development, which rose dramatically. Shot-selection became more inefficient (further inside of an improving shot-development frontier) over time. Shooter-development is the general ability of players to shoot from randomly-selected locations, and shot-selection is the ability of players to select hot-spot locations.



2013-14 through 2025-26 Shot Selection Data Summary

Zone	Attempts	Share	θ	Exp. Pts.	SD
above break 3	755,379	27.2%	0.171	1.220	1.444
0-3 ft	736,316	26.5%	0.219	1.549	0.970
3-10 ft	501,631	18.1%	0.223	1.070	1.014
16 ft-arc	280,111	10.1%	0.214	1.010	1.004
10-16 ft	271,812	9.8%	0.220	1.062	1.013
corner 3	230,492	8.3%	0.169	1.331	1.474
Total FGA	2,775,740	100.0%	0.202	1.253	1.177
zone-share = $-0.071 + 0.223 \text{ Exp.Pts.} - 0.028 \text{ SD}$, $R^2 = 0.273$					
zone-share = $-0.095 + 0.217 \text{ Exp.Pts.}$, $R^2 = 0.267$					
zone-share = $+0.153 + 0.012 \text{ SD}$, $R^2 = 0.001$					

2025-26 Shot Selection Data Summary

Zone	Attempts	Share	θ	Exp. Pts.	SD
above break 3	68,675	31.3%	0.124	1.175	1.441
0-3 ft	50,758	23.1%	0.250	1.662	0.941
3-10 ft	45,887	20.9%	0.250	1.171	1.024
corner 3	22,439	10.2%	0.124	1.284	1.470
10-16 ft	20,887	9.5%	0.250	1.137	1.021
16 ft-arc	10,835	4.9%	0.250	1.045	1.006
Total FGA	219,481	100.0%	0.198	1.288	1.201
zone-share = $-0.238 + 0.202 \text{ Exp.Pts.} + 0.133 \text{ SD}$, $R^2 = 0.245$					
zone-share = $-0.055 + 0.178 \text{ Exp.Pts.}$, $R^2 = 0.148$					
zone-share = $+0.049 + 0.102 \text{ SD}$, $R^2 = 0.059$					

Figure 2: Left: the league's thirteen-season path, pale to red, against the 2013-14 and 2025-26 frontiers, with the 2025-26 portfolio starred and improvement directions A, B, C; the shaded region holds every allocation beating it on both axes. Right: share of attempts by zone, from the 2025-26 league (dot) to balanced solution C (arrowhead). Below, attempt-weighted: theta is shot-pursuant free-throw value per attempt, expected points adds it to field-goal value, SD is scoring standard deviation per attempt.

3. Results

Shooter-development, estimated by League conversion-rate across floor-zones without weighting for location-conditional volume, **rose by 9.32 ppg from 2013-26**. This pushed the efficiency frontier out accordingly. The League moved up and to the right (from dot-to-star in left panel of Fig.2), accepting more risk for additional points. However, shot-selection inefficiency, vertical-distance within season-level frontier, **rose from 4.08 points to 6.68 points** (Fig.1), 2.6-ppg more inefficient. Moreover, team-game shot-portfolios gained 0.70 additional points of standard-deviation (scoring-risk). **Shooting became better, but only because of development and in spite of growing shot-selection inefficiency.** At-the-rim returns 1.662 points per attempt at 0.941 standard deviation, the highest-return and lowest risk zone; Above-the-break 3PAs return 1.175 at 1.441 on 31.3-percent of FGAs against 23.1-percent at-the-rim. The latter zone-share has fallen over the sample despite risk-return dominance. and the former has risen. Fig.2 shows that, from 2013-26 overall, NBA teams increased shot-volume with greater zone-return but shot-volume didn't respond to zone-risk. By 2025-26, zone-volume increased less strongly in zone-return than previously, and NBA players were **more likely to seek high-risk over low-risk zones for the same return!** This is what came of a data revolution. Teams became worse at finding hot-return zones, and more actively sought hot-risk zones. Consequently, risk and return now explain less of the variation in zone-level shot-volume than in 2013. These regressions demonstrate substantial shot-selection efficiency *regression!*

4. Conclusion

The modern NBA is thus nowhere near its shot-selection frontier. Analytics staffs have made their presence felt strictly in the first order, ignoring the *risk-level* of shot-selection portfolios while miscounting return. Wall Street quant-advisers do not recommend an equity-heavy portfolio unless the premium justifies the risk. NBA analytic-advisers have gone wing-extended long on the long-ball and short at-the-rim. Modern shot-selection does not miss the efficiency mark; it misses the entire spectrum of marks, and badly. The empress or emperor may have clothes, but lacks spatial efficiency entirely. Given herding, it *pervades*. In a closed, revenue-cartelized League, moreover, it has worsened in the very era launched to eradicate it. With this level of inefficiency in a bottom-line setting like Wall Street, for example, there would be no shirts at this point at the very least.